

# RevIN Experiment 1: Normalization Components

## 1 Theoretical overview and goal

Global standardization applies training-wide statistics. Reversible instance normalization instead uses the current look-back:

$$\tilde{x} = \alpha \frac{x - \mu_x}{\sigma_x} + \beta, \quad \hat{y} = \sigma_x \frac{f_\theta(\tilde{x}) - \beta}{\alpha} + \mu_x.$$

Non-affine instance normalization fixes  $(\alpha, \beta) = (1, 0)$ . It removes window-specific offset and scale, which can reduce temporal and cross-user shift, but it also discards potentially predictive level information.

**Experimental goal.** Isolate the contribution of global normalization, reversible per-instance normalization, and RevIN’s learnable affine parameters under the same data-space training loss.

## 2 Implementation details

The primary MSE-trained comparison is `none_mse`, `standard_mse`, `instance_mse`, and `revin_mse`. Standard statistics are fitted on training data only. Instance and RevIN statistics are computed from each look-back and inverted after forecasting; all reported MSEs are in original data space.

The complete sweep uses ETTh1, Electricity, Traffic, Solar, Weather, and Exchange Rate; seven  $(L, H)$  settings; DLinear and PatchTST; five seeds; batch size 256; learning rate  $10^{-5}$ ; and 10,000 epochs. The long-horizon setting is (504, 504). Validation and logging run every 1,000 optimizer steps, while evaluation windows use a horizon-sized stride. Training windows are sampled at random over users and dates. Test mode uses Electricity/Solar, two settings, PatchTST, seeds 1–2, and a short 20-epoch gate.

**Implementation:** `src/utils/models.py`

**Launcher:** `revin.slurm`

Run isolated test mode first. The full Cartesian product must then be sharded with `RUN_MODE=train` and followed by a dependent `RUN_MODE=tables` job. One table is produced per model and test split, with seed mean  $\pm$  sample standard deviation and an explicit row multiplier  $\times 10^m$ . JSON summaries also retain the variance of per-point losses.

## 3 Interpretation

The none–standard contrast measures global conditioning. Standard–instance measures the value of sample-specific alignment. Instance–RevIN isolates the affine parameters; it should not be described as the effect of instance normalization itself.